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Measurement-Oriented Microscopy Reconstruction for Cross-Magnification Micro-Displacement Inspection

Yen Hsuan Chiang, Liang-Chia Chen*

Department of Mechanical Engineering, National Taiwan University

No. 1, Sec. 4, Roosevelt Rd., Taipei 10617, Taiwan. *Correspondence: lchen@ntu.edu.tw

Introduction

Micro-displacement inspection requires image features that can be located consistently when the objective changes. Optical blur and pixel sampling affect the boundary signals used to estimate position.

MOSR combines calibrated optical guidance, anti-aliased multiscale features [1,2] and bounded correction on the original pixel grid. One fixed set of reconstruction weights serves 20× and 50×. We test its measurement benefit using the same circle-center procedure for Raw, fixed Wiener and MOSR.

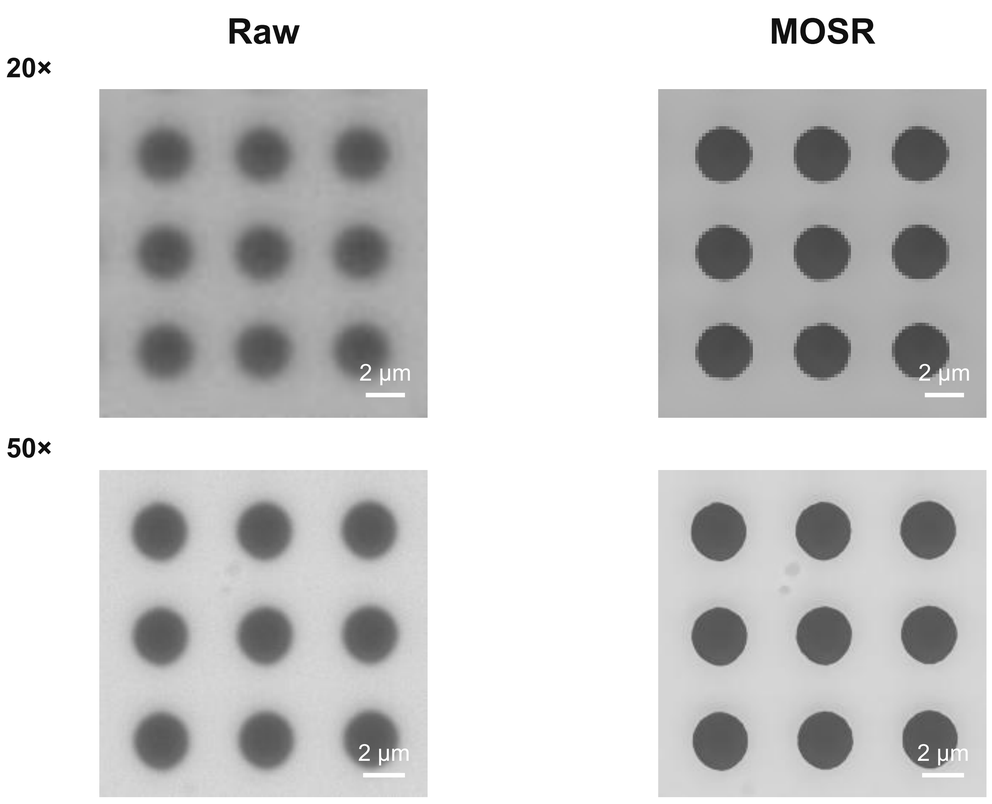


Fig. 1. Corresponding Raw and MOSR fields at both objectives. Paired images share the crop and grayscale range. Scale bars: 2 μm.

Methods

Optically guided reconstruction on the native grid

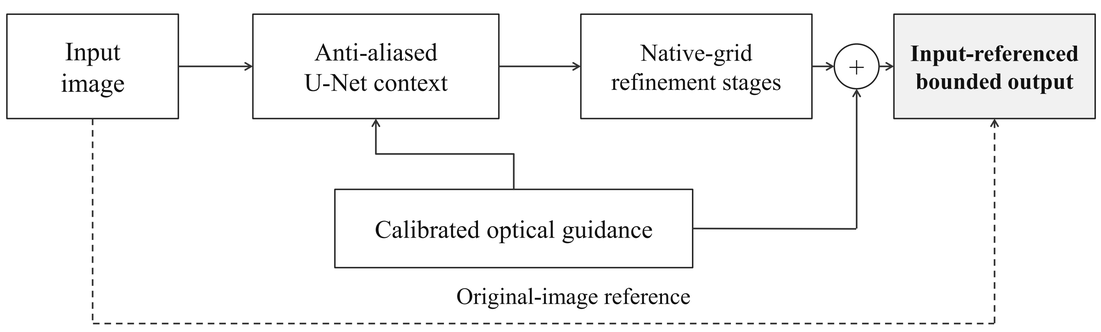


Fig. 2. Functional design of MOSR. Calibrated guidance addresses optical variation; anti-aliased features target sampling sensitivity. Refinement and bounded output retain the native image grid.

Common readout for Raw, fixed Wiener and MOSR

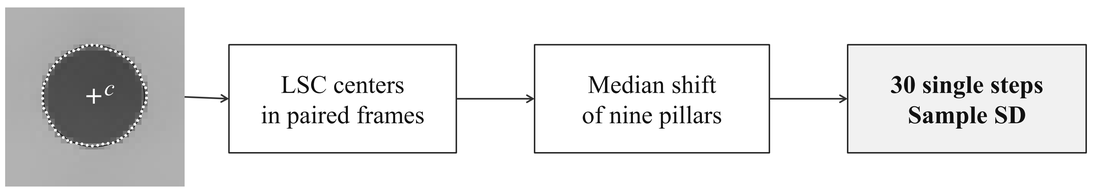


Fig. 3. Coarse contours [3] initialize boundary extraction. Circle fits [4] provide the centers used for displacement. The inset shows measured points and a fitted center.

TABLE I. EVALUATION AT A 1 μm NOMINAL STEP

Objective	Acquisition	Steps	μm/pixel
20×	31-frame evaluation scan	30	0.1725
50×	30 locked adjacent pairs	30	0.0690

Centers come from maximum-gradient boundary points and geometric LSC fits. Nine-pillar median shifts use the acquisition direction. Each condition reports the sample SD of 30 steps (denominator 29).

Results

At a common nominal step of 1 μm, MOSR gives the lowest observed Step SD in both evaluations. Each comparison uses the same images and measurement settings for all three methods.

20×, 1 μm: 30 single steps

57.2%

Lower Step SD relative to Raw

MOSR Step SD 4.068 nm

Step SD (nm)	Raw	Wiener	MOSR
	9.506	5.794	**4.068**

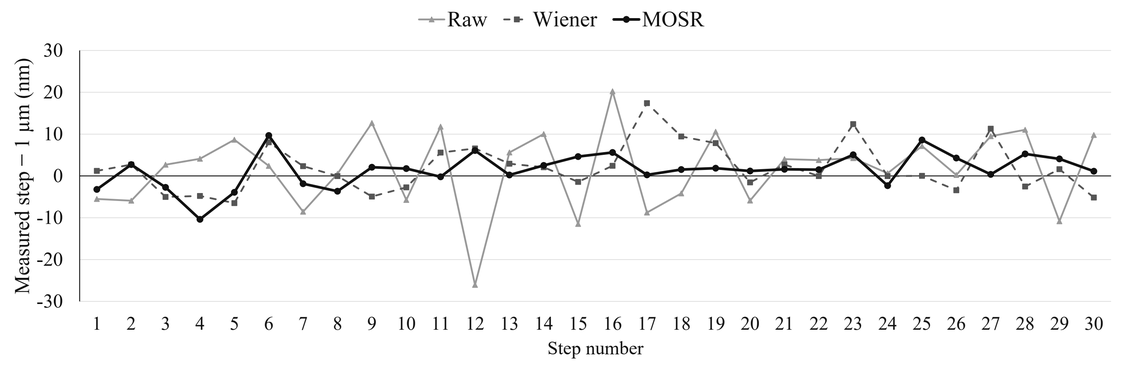


Fig. 4. All 30 single-step displacements at 20×. Curves use measured step minus 1 μm, with the same axis limits as the 50× evaluation.

At 20×, the mean step changes by only 0.10 nm while Step SD decreases by 5.44 nm. This pattern is consistent with more stable boundary-based localization.

50×, 1 μm: 30 single steps

23.5%

Lower Step SD relative to Raw

MOSR Step SD 4.542 nm

Step SD (nm)	Raw	Wiener	MOSR
	5.938	4.722	**4.542**

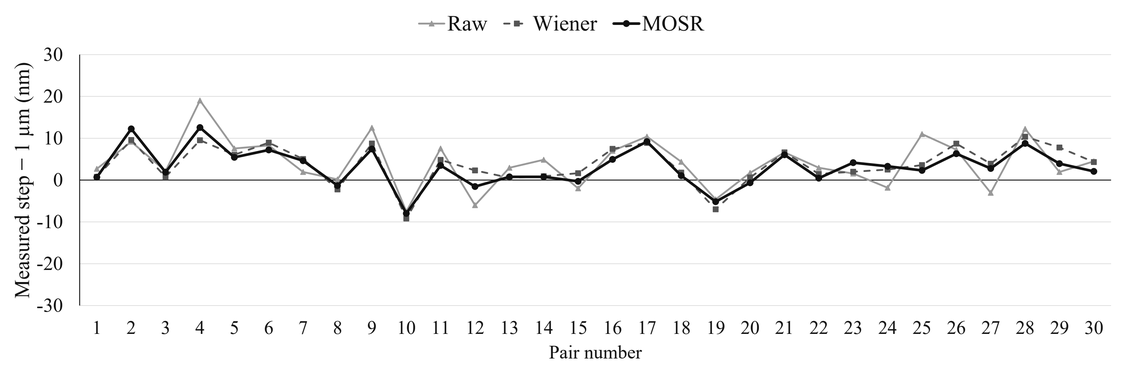


Fig. 5. All 30 locked single-step displacements at 50×. MOSR reaches 4.542 nm Step SD with the same learned weights and the corresponding optical calibration.

Conclusions

With shared reconstruction weights and objective-specific optical guidance, MOSR reduces single-step variation by 57.2% at 20× and 23.5% at 50×. The observed reductions under a common center readout support MOSR for cross-magnification micro-displacement inspection.

Bibliography

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